

1. [6 marks]

4.2.3 (2015:31)

- (a) Write a balanced ionic equation to represent the reaction described below. Include all state symbols.

0.100 mol L⁻¹ aqueous solutions of silver nitrate and potassium carbonate are mixed. [3]

- (b) Describe a chemical test that can be used to distinguish between magnesium solid and cobalt solid. State the observations expected for each of the solids when tested. [3]

Chemical test

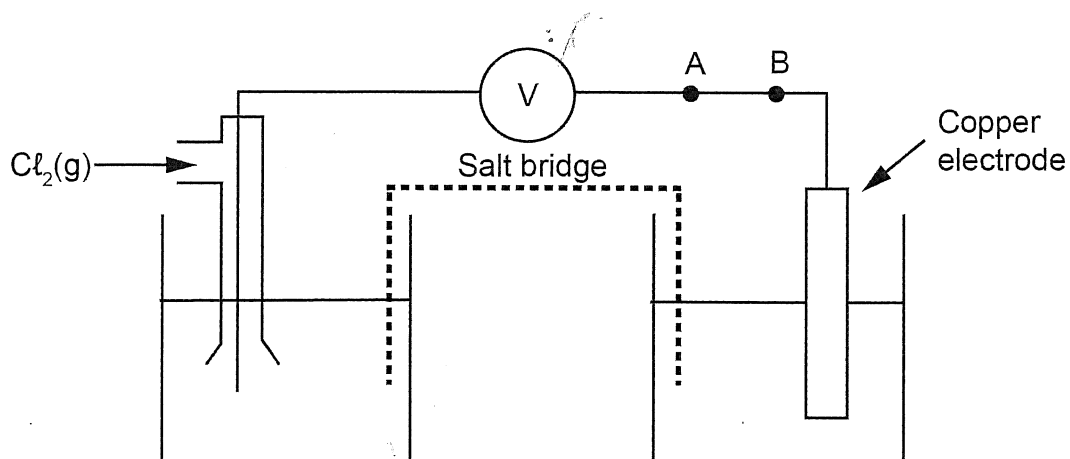
Observation with magnesium solid

Observation with cobalt solid

2. [6 marks]

4.2.6, 4.2.9 (2015:35)

The following galvanic cell was set up under standard conditions.



1 mol L⁻¹ sodium chloride solution 1 mol L⁻¹ copper(II) nitrate solution

- (a) Draw an **arrow** between **A** and **B** on the diagram to indicate the direction of electron flow. [1]
- (b) Write a balanced equation to represent the overall reaction occurring in this cell. [2]

- (c) State the reason for the reactants being kept in separate half-cells. [1]

- (d) State the observation predicted to occur in the Cl₂/NaCl half-cell. [1]

- (e) Predict a metal/metal ion cell that could be used in place of the Cu/Cu²⁺ cell to give a higher emf (volts). [1]

3. [16 marks]

Sulfur compounds in sewage and industrial processes can cause problems due to their odours, often because they eventually form dihydrogen sulfide gas, also known as rotten egg gas.

One class of sulfur compounds that need to be removed from sewage is the thiosulfates. One step in their removal is the reaction of tetrathionate ions, $\text{S}_4\text{O}_6^{2-}$, with hydrogen peroxide, H_2O_2 . The tetrathionate produces trithionate ions, $\text{S}_3\text{O}_6^{2-}$, and sulfate ions.

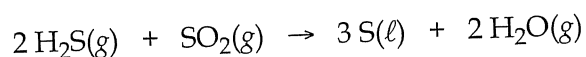
- (a) (i) Complete the table below by writing balanced half-equations and the final redox equation for the reaction of tetrathionate and hydrogen peroxide. [6]

Half-equation one
$\text{S}_4\text{O}_6^{2-} \rightarrow \text{S}_3\text{O}_6^{2-} + \text{SO}_4^{2-}$
Half-equation two
$\text{H}_2\text{O}_2 \rightarrow$
Redox

- (ii) Which substance is being oxidised? [1]

In some industrial processes, dihydrogen sulfide is a waste product. If the quantity of gas is large, then it becomes economical to extract for other production processes. The Claus process is one such example where dihydrogen sulfide is used to produce elemental sulfur.

One of the key reactions occurring in the process is the Claus reaction represented by the equation shown below.



Modern Claus plants, using three catalytic converters, may achieve up to 99.8% conversion but typically the conversion of dihydrogen sulfide to sulfur is between 95 and 97%.

dihydrogen sulfide (H_2S)	mass = 19.5 kg
sulfur dioxide (SO_2)	pressure = 68.3 kPa temperature = 791 °C volume = 43.4 kL

- (b) Calculate the maximum mass of sulfur that can be formed if the process is 96.8% efficient. Express your answer to **three** significant figures. [9]

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4. [6 marks]

4.2.6, 4.2.7 (2016:26)

Galvanic cells and electrolytic cells are often constructed in the laboratory.

- (a) List **four** characteristics or components that these two types of cells have in common with each other. [4]

One: _____

Two: _____

Three: _____

Four: _____

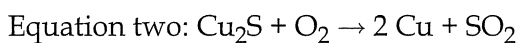
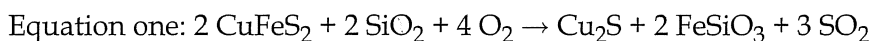
- (b) List **two** characteristics or components that can be used to distinguish between the two types of cells. State the characteristic or component for each cell. [2]

One: _____

Two: _____

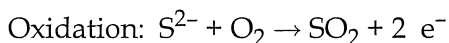
- (i) The first process occurs in a furnace where the chalcopyrite is converted to 'blister copper', which is approximately 98% pure due to impurities such as sand.
- (ii) The second process occurs in an electrolytic cell where the 'blister copper' undergoes electrolysis to produce copper at or above 99.9% purity.

Below are the equations that represent the main processes occurring in the blast furnace.



(a) Equation two can be represented as half equations. Write the reduction half equation. [1]

Reduction: _____



In the electrolytic cell, the copper produced from the blast furnace is purified.

(b) Explain the electrolytic process used to purify copper. Include:

- a brief overview of the process
- a labelled diagram of the electrolytic cell
- the relevant oxidation and reduction half equations
- a discussion of impurities and how they are separated from the copper. [10]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

6. [16 marks]

Use the following information about bleaching to answer the questions that follow.

Many chlorine-based compounds, such as sodium hypochlorite (NaOCl), chlorine (Cl_2) and chlorine dioxide (ClO_2), are used as bleaches in household cleaning products and for industrial processes.

Their uses include:

- removing colour (for example, stain removal from clothes)
- whitening paper pulp in the process of making paper
- sterilising substances (for example, swimming pool water).

These compounds act by oxidising the compounds with which they come into contact. When chlorine gas is used for bleaching, the active ingredient is hypochlorous acid (HOCl). This is produced by reaction of the chlorine gas with water. Hydrochloric acid is also produced in the reaction.

To increase the amount of hypochlorous acid produced in this reaction, the water through which the chlorine is bubbled is usually made alkaline by the addition of a small amount of hydroxide ions. Chlorine-based bleaches react well at room temperatures.

A disadvantage of chlorine bleaches is the potential for highly poisonous dioxins to be produced by reaction with organic compounds. Peroxide bleaches are environmentally more acceptable because they produce only oxygen and water.

Hydrogen peroxide is a liquid, but sodium percarbonate ($2 \text{Na}_2\text{CO}_3 \cdot 3 \text{H}_2\text{O}_2$) and sodium perborate ($\text{NaBO}_3 \cdot \text{H}_2\text{O}$) are solid peroxide bleaches that release hydrogen peroxide when dissolved in water. A disadvantage of peroxide bleaches is the need for high temperatures for them to react.

The development of molecules known as tetra-amido macrocyclic ligand-activators (TAMs) that function as catalysts has enabled the hydrogen peroxide bleaching reaction to occur at much lower temperatures.

- (a) Write the balanced equation for the reaction of chlorine gas with water. [1]

- (b) Explain briefly how the addition of hydroxide ions to the water through which the chlorine is bubbled will increase the amount of hypochlorous acid produced. [3]

- (c) Give the oxidation state of chlorine in hypochlorous acid and the oxidation state of chlorine in hydrochloric acid. [2]

Oxidation state of Cl in hypochlorous acid	Oxidation state of Cl in hydrochloric acid

- (d) Identify the species oxidised and the species reduced when chlorine gas reacts with water. [2]

Species oxidised	Species reduced

- (e) Write the half-equation for the reduction of hypochlorous acid to chloride ion. [1]

- (f) For every one mole of formula units of sodium percarbonate, how many mole of hydrogen peroxide are released when it is dissolved in water? [1]

- (g) Compare the activation energy for oxidation reactions involving chlorine-based bleaches to those using peroxide-based bleaches (in the absence of catalysts). Explain the reasons for your answer. [2]

- (h) Draw a fully labelled energy profile diagram showing the progress of the decomposition of hydrogen peroxide with and without TAML molecules. [4]

The equation for the reaction is $2 \text{H}_2\text{O}_2 \rightarrow 2 \text{H}_2\text{O} + \text{O}_2 + \text{energy}$.

7. [13 marks]

3.2.9, 4.2.3 (2017:26)

A limited amount of sodium metal was added to a beaker of distilled water containing a few drops of phenolphthalein.

- (a) List three changes that would be observed. [3]

One: _____

Two: _____

Three: _____

Write the ionic equation for any reaction involving both sodium and water. Include all state symbols. [3]

Excess propene gas was bubbled through an aqueous bromine solution.

- (b) (i) Identify, by name or formula, any new substance/s produced. [1]

- (ii) Write descriptions of the substances before and after mixing. [2]

Before _____

After _____

A thick strip of lead metal was immersed into a small beaker containing a solution of 1.00 mol L^{-1} iron(III) nitrate.

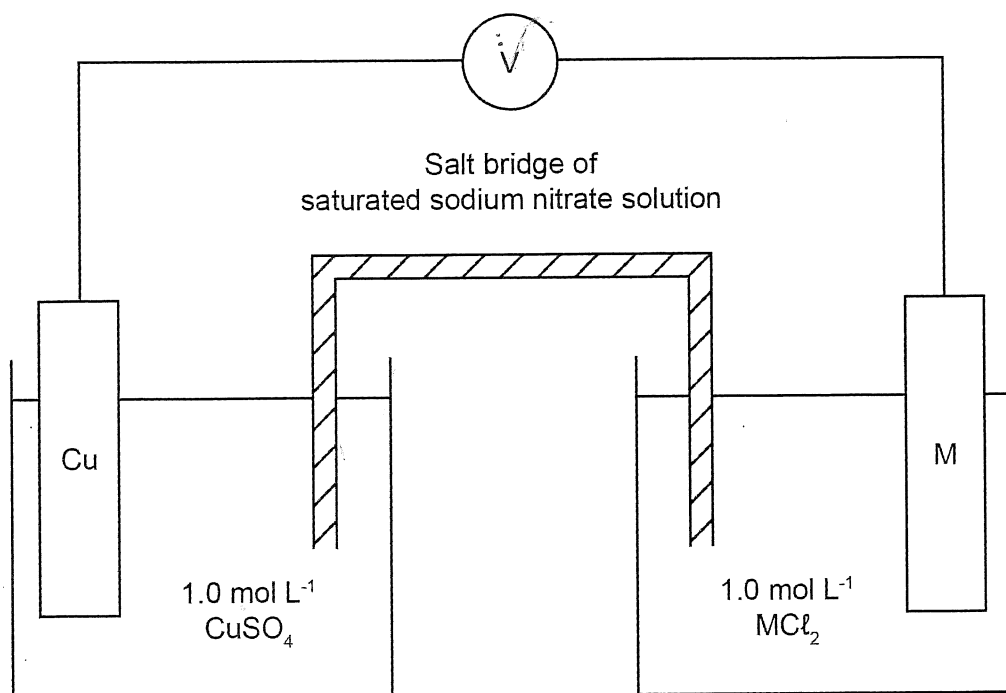
- (c) (i) Identify, by name or formula, any new substance/s produced. [1]

- (ii) List all observations that would be made for any reaction, describing clearly the substances before and on completion of any reaction. [3]

8. [7 marks]

4.2.6, 4.2.8 (2017:34)

The diagram below represents a simple galvanic cell set up at 25.0 °C.



One electrode/electrolyte pair is Cu/Cu^{2+} . The other electrode is of an unknown metal, represented as M/M^{2+} . It was observed, that over time, the unknown metal electrode reduced in size and the solution remained colourless.

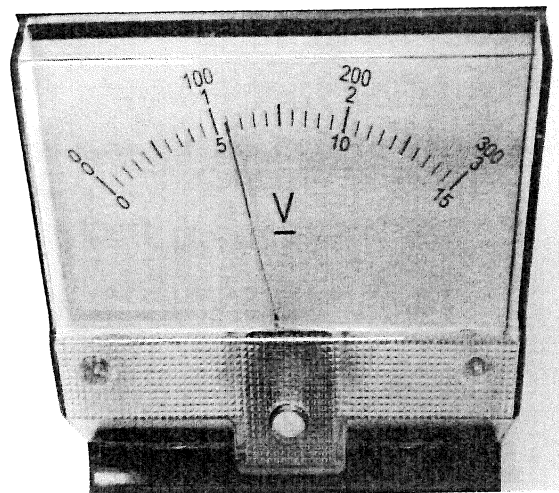
- (a) Write a chemical equation to show the reaction at the anode of the cell. [1]

- (b) List **two** observations that would be expected in the Cu/Cu^{2+} cell. [2]

One: _____

Two: _____

Below is a photograph of the voltmeter attached to the diagram of the cell on the previous page. There are three scales on the voltmeter. The scale being used is the one with the range from 0 to 3 volts.



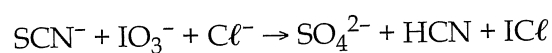
- (c) (i) To the appropriate degree of accuracy, what is the reading on the voltmeter? [1]

- (ii) Using the voltmeter reading and other relevant information, predict the identity of the unknown metal. Clear reasoning, including a calculation, must be provided. [3]

9. [6 marks]

4.2.2 (2017:27)

Balance the following redox equation by determining and then combining the oxidation and reduction half-equations. State symbols are not required.



Oxidation half-equation

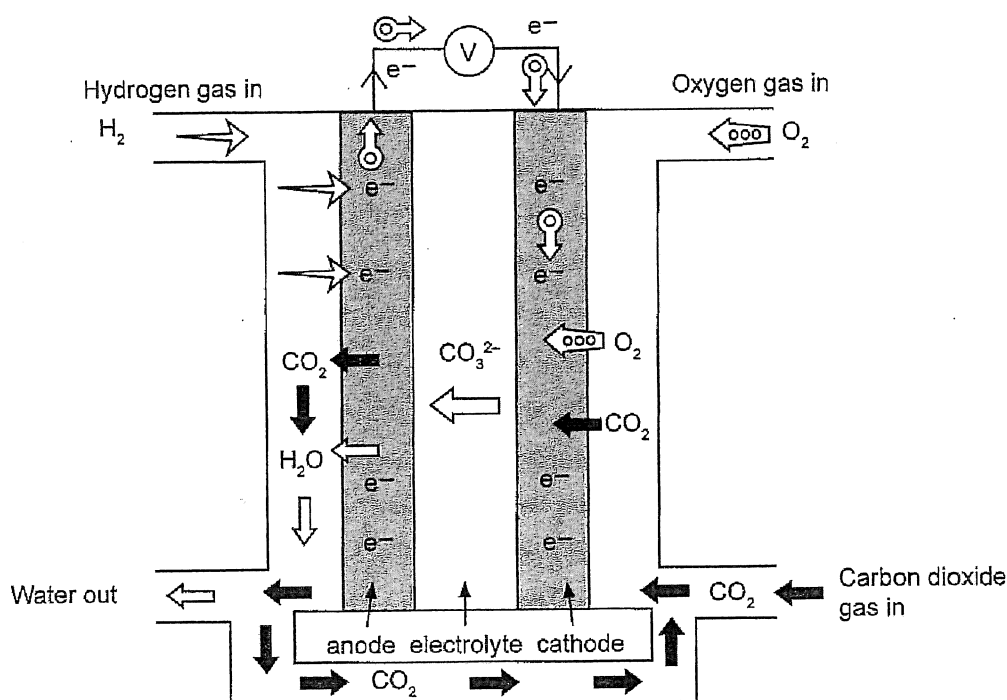
Reduction half-equation

Overall equation

10. [8 marks]

4.1, 4.2.2, 4.2.6 (2018:28)

An example of a galvanic cell is the molten carbonate fuel cell, represented in the diagram below. As this cell operates, hydrogen gas is reacted with the carbonate ion at the anode, while oxygen gas reacts with carbon dioxide gas at the cathode. The carbon dioxide gas is re-used.



- (a) Write the half-equation to show the reaction at the electrode at which oxidation occurs. [3]

- (b) Write the overall equation for the current-producing reaction. [3]

- (c) State **two** reasons why this fuel cell is a more environmentally-friendly alternative to the internal combustion engine. [2]

One: _____

Two: _____

11. [7 marks]

Calcium hypochlorite and hydrochloric acid react according to the equation shown below.



In this reaction, the chlorine in calcium hypochlorite and the chloride from the hydrochloric acid are both converted to chlorine gas.

(a) State the oxidation number for chlorine in:

[2]

- calcium hypochlorite, $\text{Ca}(\text{OCl})_2$
- hydrochloric acid, HCl

calcium hypochlorite

hydrochloric acid

Chlorine gas is produced by the oxidation of one of these substances and the reduction of the other.

(b) Write the **two** half-equations showing how chlorine gas is produced from both substances. [5]

Oxidation half-equation

Reduction half-equation

12. [8 marks]

4.2.3 (2020:27)

Write balanced equations for any reactions occurring between the following substances and describe the observation(s).

If there is no reaction, write 'no reaction' for the equation and if there is no change observed, write 'no visible reaction' for the observations. Where applicable, use the colours stated in the Chemistry Data booklet.

Iron filings and dilute hydrochloric acid

Equation
Observation(s)

Chromium(III) nitrate solution and magnesium ribbon

Equation
Observation(s)

Potassium chloride solution and bromine water

Equation
Observation(s)

13. [16 marks]

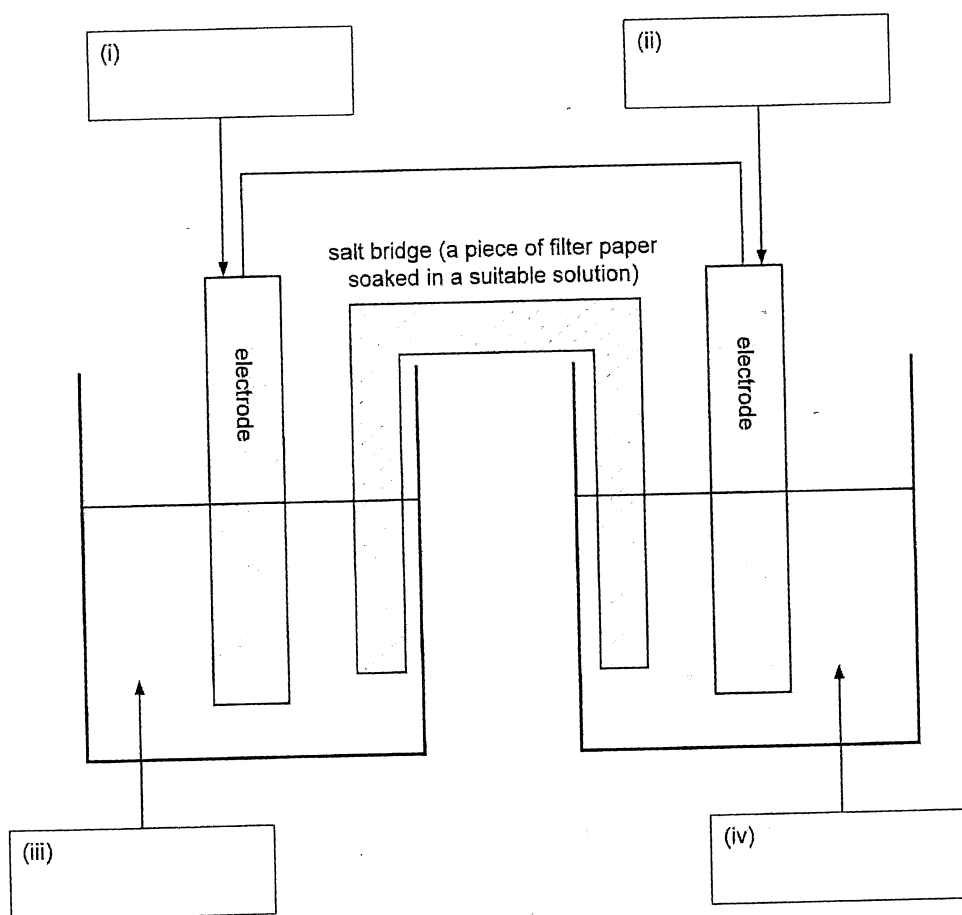
4.2.6, 4.2.7, 4.2.8, 4.2.9 (2020:36)

A student was asked to build a functioning galvanic cell, having been provided with all of the required hardware plus the following substances:

- a piece of magnesium measuring 1 mm by 2 cm by 6 cm
- a piece of copper measuring 1 mm by 2 cm by 6 cm
- a 6 cm long graphite (carbon) rod with a diameter of 1 cm
- 1.0 mol L^{-1} sodium carbonate solution
- 1.0 mol L^{-1} magnesium sulfate solution
- 1.0 mol L^{-1} copper(II) sulfate solution.

There was no requirement for the student to use all of these substances.

- (a) A partially-labelled diagram of the galvanic cell built by the student is shown below. What substances should the student have used in the parts labelled (i) to (iv) to build a functioning galvanic cell? Write the names of these substances in the boxes provided. [4]



- (b) Add arrows to the diagram in part (a) to show the direction of movement of electrons through the external circuit. [1]
- (c) Write the half-equations for the reactions occurring at the anode and the cathode in the student's galvanic cell. [4]

Anode half-equation _____

Cathode half-equation _____

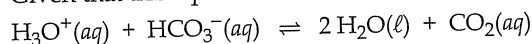
- (d) Calculate the electrical potential difference of the student's galvanic cell. Assume standard conditions. Include appropriate units in your answer. [2]

- (e) Galvanic cells, such as the one shown in the diagram, need a salt bridge.

- (i) State why galvanic cells need a salt bridge. [1]

- (ii) Describe, with reference to ion movement, how the salt bridge in a galvanic cell works. Also state why ion movement occurs as you have described. [4]

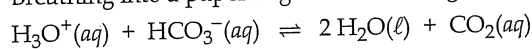
- e) That the conjugate pair that make up the buffer (hydrogencarbonate ion and carbonic acid) are of equal and high concentration.
- f) Using Le Châtelier's Principle means we do not discuss forward and backward rates and collisions but instead refer to overcoming a constraint to re-establish equilibrium.
- Given that this equilibrium existed.



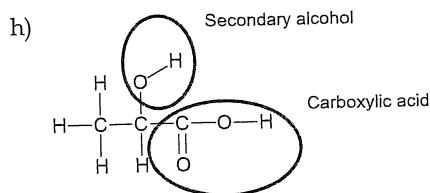
When the kidneys applied a constraint by removing hydrogencarbonate ions the equation moved to the left to re-establish equilibrium. In doing so more hydronium ions are formed. So the $[\text{H}_3\text{O}^+]$ increases and the pH decreases.

- g) This question uses the term 'explain'. The syllabus specifies that 'explaining' can only be done with collision theory. Thus LCP is not acceptable here.

Breathing into a paper bag and re-inhaling the same air increases the $[\text{CO}_2]$ in the blood.

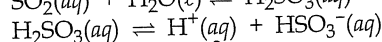
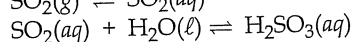
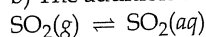


Therefore, the frequency of collisions of the products will increase and there will be a higher frequency of successful collisions. The reverse rate of reaction will increase. The forward rate initially remains the same so the position of equilibrium moves to the left. This increases $[\text{H}_3\text{O}^+]$ and the pH of the blood is lowered overcoming the pH rising effect of hyperventilation.

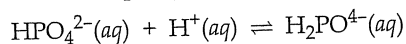


- 13.(2020:30) a) Atom that is oxidised - sulfur (oxidation number +4 $\rightarrow$ +5)
Atom that is reduced - hydrogen (oxidation number +1 $\rightarrow$ 0) in $\text{H}_2(\text{g})$

- b) The addition of $\text{SO}_2(\text{g})$ will form an acidic solution:



As $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$ is a weak acid and its conjugate, it forms a buffer. The $\text{H}^+(\text{aq})$ produced above will react with the $\text{HPO}_4^{2-}(\text{aq})$ base in the buffer.



This reaction will consume most of the $\text{H}^+(\text{aq})$

$\therefore$ the $[\text{H}^+(\text{aq})]$ will increase only slightly so the pH ($-\log[\text{H}^+]$) will fall insignificantly.

Chapter 9: Oxidation-Reduction

- 1.(2015:31) a) $2 \text{Ag}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{Ag}_2\text{CO}_3(\text{s})$
b) To a sample of each metal in different test tubes add 0.100 mol L⁻¹ solution of chromium nitrate solution.
Magnesium: green solution decolourises
Cobalt: no visible reaction
- 2.(2015:35) a) arrow to left
b) $\text{Cl}_2(\text{g}) + \text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2 \text{Cl}^-(\text{aq})$. Always use ionic equations.
c) The electrons that are transferred can pass through an external circuit (rather than being transferred through direct contact)
d) Greenish yellow gas disappears
e) Any metal/metal ion below 0.34 V eg Zn/Zn^{2+} : must not be K, Na, Ca, Sr, Ba as they react with cold water.
- 3.(2015:37) a) i) Half-equation one:
$$4 \text{H}_2\text{O}(\ell) + \text{S}_4\text{O}_6^{2-}(\text{aq}) \rightarrow \text{S}_3\text{O}_6^{2-}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 8 \text{H}^+(\text{aq}) + 6 \text{e}^-$$

Half-equation two:
$$2 \text{e}^- + 2 \text{H}^+(\text{aq}) + \text{H}_2\text{O}_2(\text{aq}) \rightarrow 2 \text{H}_2\text{O}(\ell)$$

Redox:
$$\text{S}_4\text{O}_6^{2-}(\text{aq}) + 3 \text{H}_2\text{O}_2(\text{aq}) \rightarrow \text{S}_3\text{O}_6^{2-}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) + 2 \text{H}_2\text{O}(\ell) + 2 \text{H}^+(\text{aq})$$

ii) tetrathionate or $\text{S}_4\text{O}_6^{2-}$
b) $n(\text{H}_2\text{S}) = 19.5 \times 10^3 / 34.086 = 572.08 \text{ mol}$
 $n(\text{SO}_2) = PV/RT = (68.3 \times 43.4 \times 1000) / (8.314 \times 1064.15) = 335.04 \text{ mol}$
 $n(\text{H}_2\text{S}) \text{ required} = 2 n(\text{SO}_2) = 2 \times 335.04 = 670.08 \text{ mol}$
There is insufficient H_2S (572 mol) for all SO_2 to react, the SO_2 is in excess and so H_2S is the limiting reagent
 $n(\text{S}) = 3/2 \times n(\text{H}_2\text{S}) = 3/2 \times 572.08 = 858.12 \text{ mol}$
 $m(\text{S}) = 858.12 \times 32.07 = 27\,519.9 \text{ g} = 27.5 \text{ kg}$ or $2.75 \times 10^4 \text{ g}$
For 96.8% efficiency: $m(\text{S}) = 27.5 \times 0.968 = 26.6 \text{ kg}$

4.(2016:26) a)

- both have an external wire for electron flow
- both have a cathode and an anode
- positive ions flow towards the cathode
- electrons flow through an external circuit from anode to cathode
- an electrolyte/liquid/solution is needed
- one substance is oxidised and the other reduced
- oxidation occurs at the anode and reduction occurs at the cathode
- both have electrodes

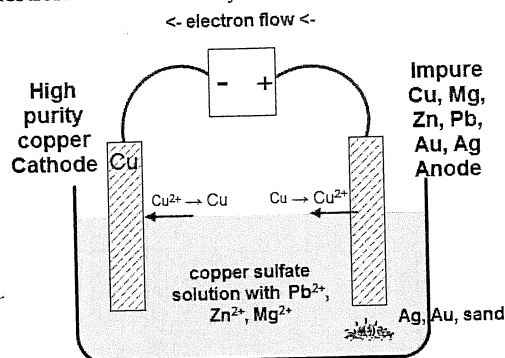
b)

- a galvanic cell is linked to a load while an electrolytic cell is linked to a power supply
- galvanic cells are spontaneous while electrolytic are forced
- there is only one beaker (cell) in an electrolytic cell and two half beakers (cells) in a galvanic cell
- galvanic has a salt bridge while electrolytic cell does not
- the galvanic cell has a spontaneous reaction while an electrolytic cell requires an emf to be applied to force a reaction (non-spontaneous)
- the electrolytic cell has an external emf applied (battery/power source) while the galvanic cell produces the emf
- galvanic cell has a positive E_o while electrolytic the E_o is negative
- in a galvanic cell, chemical energy is converted to electrical energy and in an electrolytic cell, electrical energy is converted to chemical. 1-2.

5.(2016:40) a) $\text{Cu}^+(aq) + e^- \rightarrow \text{Cu}(s)$

b) This question is taken directly from the syllabus. It is required knowledge and so this is fairly straight forward.

Diagram:



The electrolytic refining of copper involves the dissolution of an impure blister copper sample (used as the anode) in a copper sulfate solution, and the subsequent deposition of copper ions as solid on the pure copper electrode used as the cathode.

A potential difference is applied across the electrodes giving a positive charge to the anode and a negative charge to the cathode.

This induces an otherwise non-spontaneous redox reaction in which the copper is oxidised and reduced.

Oxidation equation: $\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2e^-$

Reduction equation: $\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$

The electrons flow from the ionised blister copper via the external circuit to the copper cathode.

The dissolved copper ions will then flow towards the negatively charged cathode due to electrostatic attraction, where they are deposited as metallic solid.

Impurities in the original blister copper sample are either dissolved preferentially in solution and stay there (such as zinc or magnesium) or alternatively drop down and form a mud (gold and silver do this) as copper is preferentially oxidised and they do not react.

The voltage is chosen so not to oxidise gold or reduce zinc etc.

6.(2016 SP:42) a) $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HOCl} + \text{HCl}$

b)

- recognition that the hydroxide will react with the H^+
- recognition that removal of H^+ will prevent reverse reaction from occurring
- recognition that the forward reaction will thus continue until close to completion thus increasing yield of HOCl

c)

Oxidation state of Cl in hypochlorous acid +1

Oxidation state of Cl in hydrochloric acid -1

d)

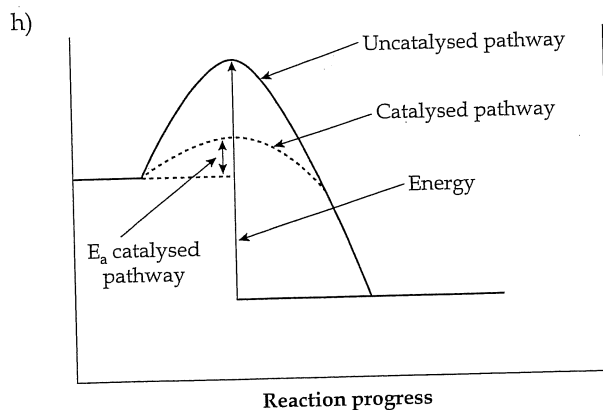
Species oxidised Cl_2

Species reduced Cl_2

e) $\text{HOCl} + \text{H}^+ + 2e^- \rightarrow \text{Cl}^- + \text{H}_2\text{O}$

f) 3 moles

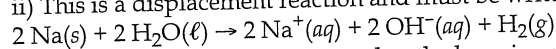
g) activation energy for oxidation of peroxides is higher than E_a of chlorine based bleaches the oxidation of peroxide based bleaches requires higher temperatures than for oxidation of chlorine based bleaches



7.(2017:26) a) i) There are many suitable observations that can be made here.

- The metal will react violently with the water
- A colourless odourless gas will be produced
- The solution will become warm
- The indicator will change to pink

ii) This is a displacement reaction and must be written as an ionic equation



b) i) 1,2-dibromopropane - remember the bromine atoms add each side of the double bond and cannot form 1,1-dibromopropane

ii) The test for a double bond using bromine water was seen in year 11. Here a colourless gas mixes with an orange solution which turns colourless. Interestingly the 1,2-dibromopropane will lie as an oily layer on the surface of the water.

c) i) By consulting the E° table you will see the Fe(III) will reduce to Fe(II) but not Fe. So we will see Pb^{2+} and Fe^{2+} form.
ii) You would see the grey lead added to brown Fe(III) solution which will turn green while the lead will partly dissolve.

8.(2017:34) a) $\text{M} \rightarrow \text{M}^{2+} + 2\text{e}^-$

b) One: The electrode becomes larger being covered by a salmon pink metal

Two: The solution becomes less blue

c) i) 1.08 V - you should read to 2 decimal places

ii) The E° for the cell is 1.08 and copper is 0.34 so the element must have an E° around 0.76. It is colourless so cannot be Cr - the other electrode must be zinc.

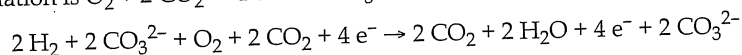
9.(2017:27) Oxidation half-equation: $\text{SCN}^- + 4\text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + \text{HCN} + 7\text{H}^+ + 6\text{e}^-$

Reduction half-equation: $\text{IO}_3^- + 6\text{H}^+ + 4\text{e}^- + \text{Cl}^- \rightarrow \text{ICl} + 3\text{H}_2\text{O}$

Balanced Equation: $2\text{SCN}^- + 3\text{IO}_3^- + 3\text{Cl}^- + 4\text{H}^+ \rightarrow 2\text{SO}_4^{2-} + 2\text{HCN} + 3\text{ICl} + \text{H}_2\text{O}$

10.(2018:28) a) Oxidation occurs at the anode: $\text{H}_2 + \text{CO}_3^{2-} \rightarrow \text{CO}_2 + \text{H}_2\text{O} + 2\text{e}^-$.

b) The reduction equation is $\text{O}_2 + 2\text{CO}_2 + 4\text{e}^- \rightarrow 2\text{CO}_3^{2-}$ so when we add the two equations we get

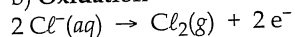


Which simplifies to $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.

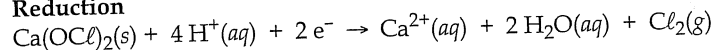
c) It reuses the carbon dioxide so has no greenhouse emissions. It produces no oxides of sulfur or nitrogen.

11.(2019:28) a) • calcium hypochlorite, $\text{Cl} = +1$
• hydrochloric acid, $\text{Cl} = -1$

b) **Oxidation**



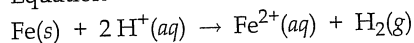
Reduction



12.(2020:27)

Iron filings and dilute hydrochloric acid

Equation

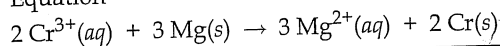


Observation(s)

A grey solid is added to a clear colourless liquid. A colourless odourless gas is formed and the solution becomes pale green.

Chromium(III) nitrate solution and magnesium ribbon

Equation



Observation(s)

A grey solid is added to a deep green solution. The grey solid dissolves, a grey solid forms and the deep green solution fades. This is an example of a metal displacement reaction because Mg is a more active reductant than Cr.

Potassium chloride solution and bromine water

Equation

'no reaction' – bromine cannot spontaneously displace chlorine.

Observation(s)

'no visible reaction' – an orange solution was added to a colourless solution and the mixture remained orange.

13. (2020:36) a) To be a functioning galvanic cell it must be spontaneous – the E° must be positive with 1 mol L⁻¹ solutions.

So magnesium and $\text{Cu}^{2+}(\text{aq})$ meet this criterion. The cathode must not react with the $\text{Cu}^{2+}(\text{aq})$ so it must be copper or graphite. The salt bridge should be ionic and soluble but not form precipitates so a carbonate would not be a good choice. (i) magnesium (ii) copper or graphite (iii) (1.0 mol L⁻¹) magnesium sulfate (solution)

(iv) (1.0 mol L⁻¹) copper(II) sulfate (solution)

b) In the external wire an arrow from left to right: →

c) Oxidation occurs at the anode. Reduction occurs at the cathode.

Anode half-equation – $\text{Mg}(\text{s}) \rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{e}^-$

Cathode half-equation – $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$

d) Note the question asks for the units to be included

$\text{Mg}(\text{s}) \rightleftharpoons \text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \quad E^\circ = +2.36\text{V}$

$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{Cu}(\text{s}) \quad E^\circ = +0.34\text{V}$

Total +2.70 V

e) i) A salt bridge provides ions to each cell to maintain neutrality. Sometimes this is expressed as completing the circuit.

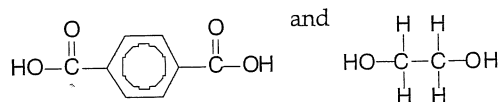
ii) In this cell, at the anode, positive $\text{Mg}^{2+}(\text{aq})$ ions are being formed (oxidation) while at the cathode, $\text{Cu}^{2+}(\text{aq})$ ions are being destroyed (reduction).

Negative ions move through the salt bridge to return the charge to zero at the anode half-cell where $\text{Mg}^{2+}(\text{aq})$ ions are being formed.

Positive ions move through the salt bridge to return the charge to zero at the cathode half-cell where $\text{Cu}^{2+}(\text{aq})$ ions are being destroyed and the solution would have been left with a negative charge.

Chapter 10: Organic Chemistry

1. (2015:32) a)



b) Water

c) Increasing the length of the polymer chain increases the dispersion forces of the polymers. The number of =O dipoles increases so the number of dipole attractions along the length of the polymer chains increases.

- This increases the magnitude of its interactions with neighbouring chains making the polymer more rigid
- More energy is required to overcome the attraction of the chains of the polymer from each other thereby raising the melting point

2. (2015:34) a)

Unknown organic compound	Functional group	Structural formula or name of the organic compound
1	alcohol	methylpropan-2-ol $\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array}$
2	aldehyde	butanal or methylpropanal $\begin{array}{c} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{H}-\text{C}- & \text{C}- & \text{C}- & \text{C} \\ & & & // \\ \text{H} & \text{H} & \text{H} & \text{O} \end{array}$ $\begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{CH}-\text{CH} \\ \\ \text{CH}_3 \end{array}$